

HDOC-29

Array Operations Using a Menu-Driven C Program

Aim

To write a menu-driven C program to perform different operations on arrays using a switch statement, loops, and conditional statements. The program counts even, odd, positive, negative, and zero elements, performs linear search, merges two arrays, and provides an exit option.

Algorithm

1. Start the program.
2. Display the menu with options to count array elements, perform linear search, merge two arrays, or exit.
3. Accept the user's choice and process it using a switch statement.
4. For Count Elements, input the array and count even, odd, positive, negative, and zero values using loops and conditions.
5. For Linear Search, input the array and search key, then traverse the array until the element is found or the array ends.
6. For Merge Arrays, input two arrays, copy their elements into a third array, and display the merged array.
7. For Exit, terminate the program.
8. For an invalid choice, display an appropriate error message.
9. Continue displaying the menu until Exit is selected.
10. Stop the program.

C Program

```
#include <stdio.h>

int main()
{
    int choice;

    do
    {
        printf("\n----- ARRAY OPERATIONS MENU ----- \n");
        printf("1. Count Even, Odd, Positive, Negative and Zero Elements\n");
        printf("2. Linear Search\n");
        printf("3. Merge Two Arrays\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
```

```

switch (choice)
{
    case 1:
    {
        int n, i;
        int even = 0, odd = 0;
        int positive = 0, negative = 0, zero = 0;

        printf("Enter size of array: ");
        scanf("%d", &n);
        int arr[n];

        printf("Enter %d elements: ", n);
        for (i = 0; i < n; i++)
            scanf("%d", &arr[i]);

        for (i = 0; i < n; i++)
        {
            if (arr[i] % 2 == 0)
                even++;
            else
                odd++;

            if (arr[i] > 0)
                positive++;
            else if (arr[i] < 0)
                negative++;
            else
                zero++;
        }

        printf("\nEven elements      : %d\n", even);
        printf("Odd elements       : %d\n", odd);
        printf("Positive elements : %d\n", positive);
        printf("Negative elements : %d\n", negative);
        printf("Zero elements      : %d\n", zero);
        break;
    }

    case 2:
    {
        int n, i, key, position = -1;
        printf("Enter size of array: ");
        scanf("%d", &n);
        int arr[n];

        printf("Enter %d elements: ", n);
        for (i = 0; i < n; i++)
            scanf("%d", &arr[i]);

        printf("Enter element to search: ");
        scanf("%d", &key);

        for (i = 0; i < n; i++)
        {
            if (arr[i] == key)
            {
                position = i;
                break;
            }
        }

        if (position != -1)
            printf("Element found at position %d.\n", position + 1);
        else
            printf("Element not found.\n");
        break;
    }
}

```

```

case 3:
{
    int n1, n2, i;
    printf("Enter size of first array: ");
    scanf("%d", &n1);
    int arr1[n1];

    printf("Enter %d elements of first array: ", n1);
    for (i = 0; i < n1; i++)
        scanf("%d", &arr1[i]);

    printf("Enter size of second array: ");
    scanf("%d", &n2);
    int arr2[n2];
    int merged[n1 + n2];

    printf("Enter %d elements of second array: ", n2);
    for (i = 0; i < n2; i++)
        scanf("%d", &arr2[i]);

    for (i = 0; i < n1; i++)
        merged[i] = arr1[i];
    for (i = 0; i < n2; i++)
        merged[n1 + i] = arr2[i];

    printf("Merged Array: ");
    for (i = 0; i < n1 + n2; i++)
        printf("%d ", merged[i]);
    printf("\n");
    break;
}

case 4:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice! Please enter a number from 1 to 4.\n");
}
} while (choice != 4);

return 0;
}

```

Test Case Table

Test Case	Choice	Input	Expected Result
1	1	n=6; 2 -3 0 4 -5 7	Even=3, Odd=3, Positive=3, Negative=2, Zero=1
2	1	n=5; 0 0 2 6 8	Even=5, Odd=0, Positive=3, Negative=0, Zero=2
3	2	10 20 30 40 50; Search 30	Element found at position 3
4	2	5 10 15 20; Search 8	Element not found
5	3	1 2 3 and 4 5 6	Merged: 1 2 3 4 5 6
6	3	10 20 and 30 40 50	Merged: 10 20 30 40 50
7	5	-	Invalid choice! Please enter a number from 1 to 4.
8	4	-	Program exits

Result

The menu-driven C program was successfully designed to perform the required array operations. It counts even, odd, positive, negative, and zero elements, performs a linear search, and merges two arrays. The use of switch-case statements, loops, conditional statements, and arrays allows all operations to be performed through a single menu-driven program.